

BSCS THESIS PROPOSAL DEFENSE MASTER STUDY GUIDE

Notre Dame of Midsayap College • College of Information Technology and Engineering (CITE)

Researchers: Daniela S. Ungab & Rogie P. Bacanto (BSCS-4) • Adviser: Ms. Doris Ann Mariano

Approved Title: Receipt or Deceit: A Cross-Architecture Analysis of Convolutional Neural Network Models in Detecting Forged Digital Transaction Receipts (ForgeGuard System)

EXECUTIVE THESIS OVERVIEW & PROBLEM STATEMENT

The Problem: Over 52.8% of retail payments in the Philippines are transacted through digital payment channels such as GCash (BSP, 2024). In municipalities like Midsayap, micro-merchants, local online sellers, and student entrepreneurs rely on downloadable receipt slips to confirm that payment has occurred. Scammers exploit this by using graphic editors (Photoshop, Canva) or automated generator apps to manipulate transaction amounts, reference numbers, or recipient names without transferring actual money (GCash Public Advisory, 2024).

The Research Gap: Human visual inspection cannot reliably distinguish sophisticated spliced text from authentic fonts, especially after images are transmitted over messaging apps like Facebook Messenger, which heavily re-compress images. Existing forensic models are either too computationally heavy for ordinary merchant hardware or fail under aggressive lossy compression. This study provides an empirical, cross-architecture comparative evaluation of three distinct CNNs (Basic CNN, MobileNetV2, ResNet50) on Error Level Analysis (ELA) features to identify the optimal accuracy-efficiency trade-off.

Scope & Delimitations (Section 1.4): The study **strictly evaluates downloadable GCash transaction receipts**. Full-screen mobile captures containing operating system status bars (carrier, battery, clock, notifications) are delimited because they introduce device-specific artifacts that do not originate from the payment platform itself.

PART I: ACADEMIC PROPOSAL GROUND TRUTH (CHAPTERS 1 & 2)

1. Statement of the Problem (SOP 1 to 5) Breakdown

The research answers five precise, sequential questions established in Section 1.2:

SOP	Official Research Question	Plain English Meaning	Measurement / Statistical Tool
SOP 1	What is the performance of the three models in classifying an image as authentic or forged using standard machine learning metrics in terms of: (1.1) Precision, (1.2) Recall, (1.3) F1-score?	How reliably does each model catch fakes without falsely accusing honest buyers?	Confusion Matrix metrics (Precision, Recall, F1-score). Recall is critical: False Negatives mean goods released for free.
SOP 2	What is the classification accuracy of the Basic CNN, ResNet50, and MobileNetV2 models in detecting: (2.1) Digitally edited transaction receipts, (2.2) Programmatically generated fake receipts?	Can the AI detect both Photoshop/Canva edits and automated web generator templates?	Subtype Accuracy evaluated separately on: (2.1) Digital edits (amount/ref/name) vs. (2.2) Synthetic template generators.
SOP 3	Is there a significant difference in the end-to-end processing time and memory load of the three models?	Does one model run significantly faster and use less RAM on standard merchant laptops/PCs?	End-to-end Latency (ms, file read to verdict) and Peak Memory (RSS in MB). Tested with Repeated-Measures ANOVA / Friedman test (H ₀₃).
SOP 4	Is there a significant difference in classification accuracy among the three models when analyzing: (4.1) Original high-resolution downloadable receipts, (4.2) Heavily compressed images?	What happens when receipts are sent via Facebook Messenger? Does compression break the model?	Two-Way (3x2) Repeated-Measures ANOVA crossing Architecture with Image Condition (H ₀₁ , H ₀₂).
SOP 5	Which of the three architectures offers the most practical balance of classification accuracy and processing efficiency on merchant-grade hardware?	Taking accuracy, speed, and memory together, which model should local sellers actually deploy?	Normalized Composite Scoring across accuracy, F1, latency, and peak memory footprint.

2. Formal Hypotheses (Section 1.9)

- **H₀₁**: There is no significant difference in classification accuracy among Basic CNN, ResNet50, and MobileNetV2 models when detecting forged GCash receipts.
- **H₀₂**: There is no significant difference in classification accuracy between original high-resolution downloadable receipts and heavily compressed images across the three models.

- **H■3:** There is no significant difference in end-to-end processing time and memory load among Basic CNN, ResNet50, and MobileNetV2 models.

3. Expert Validation: Is CCJE (Criminology) Student Validation Valid?

Panel Note: 'Can CCJE students validate the receipts? Is that methodologically valid?'

Exact Academic Answer according to Section 2.2 & Section 2.8:

- **For Technical Instrumentation Validity (NO):** The proposal officially designates **three CITE / Computing faculty members** (independent of the defense panel) to validate the technical evaluation instruments, CNN architectures, ELA algorithms, and ANOVA statistical models because digital image forensics and neural networks belong to Computer Science.
- **For Face Validity / Visual Realism Validation (YES):** CCJE (College of Criminal Justice Education / Criminology) senior students or faculty trained in **Questioned Document Examination (QDE)** can be validly utilized as human visual raters. They evaluate receipt plausibility using the study's 4-point Forgery Realism Validation Rubric (visual plausibility, layout fidelity, consistency of transaction details). If trained criminology examiners cannot distinguish the forged receipts from authentic ones with the naked eye, it provides empirical proof that human visual inspection is inadequate and that an automated CNN forensic tool is necessary.

4. Formal Empty Results Tables for Thesis Presentation & Chapters 3/4

Use these empty tables in your defense presentation slides to illustrate the experimental design:

Table A: SOP 3 - Processing Efficiency and Memory Load (Fixed Non-GPU Merchant Hardware)

Architecture	Image Condition	End-to-End Latency (ms) Mean ± SD	Peak Memory (MB) Mean ± SD	Model Params	Statistical Test (F / χ^2)	p-value	Decision on H■3
Basic CNN	Original	[]	[]	~2.1M	—	—	—
Basic CNN	Compressed	[]	[]	~2.1M	—	—	—
MobileNetV2	Original	[]	[]	~3.4M	—	—	—
MobileNetV2	Compressed	[]	[]	~3.4M	—	—	—
ResNet50	Original	[]	[]	~23.5M	—	—	—
ResNet50	Compressed	[]	[]	~23.5M	—	—	—

Table B: SOP 1 & SOP 4 - Classification Accuracy Across Conditions

Architecture	Original High-Res Accuracy (%)	Heavily Compressed Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	Delta (Δ)	Decision on H■1, H■2
Basic CNN	[]	[]	[]	[]	[]	[]	—
MobileNetV2	[]	[]	[]	[]	[]	[]	—
ResNet50	[]	[]	[]	[]	[]	[]	—

PART II: FORGEGUARD FORENSIC SYSTEM IMPLEMENTATION

1. Error Level Analysis (ELA) Physics & Forensic Basis

The Mechanism: JPEG is a lossy compression format operating in 8x8 pixel blocks. When an authentic GCash receipt is generated, all 8x8 blocks are compressed uniformly at a single quantization level. When a scammer splices an amount or edits text in Photoshop:

- The altered pixels are subjected to **double compression**, whereas untouched background pixels are not.
- ForgeGuard resaves the image at **JPEG Quality Q=90** and computes the absolute pixel difference: $\Delta = |I_{orig} - I_{resaved}|$.
- The difference is multiplied by a **15.0x scale factor**. In authentic receipts, the residual error is low and uniform. In forged receipts, modified digits exhibit high-frequency error spikes.

2. The Three CNN Architectures Compared

- **Basic CNN (~2.1M parameters):** A custom 3-block convolutional network (Conv2D → ReLU → MaxPooling → Dropout → Dense → Sigmoid). Because it is shallow, it directly extracts local spatial noise edges without abstracting them away. Achieved **92.92% standard** and **99.56% compressed** accuracy with an ultra-low latency of **22.51 ms**.
- **MobileNetV2 (~3.4M parameters):** Uses depthwise separable convolutions with inverted residual blocks. Delivers high representation power with low computational multiply-accumulate steps. Achieved **86.06% standard** and **93.42% compressed** accuracy at **215.75 ms** latency.
- **ResNet50 (~23.5M parameters):** A 50-layer deep residual network with skip connections. Designed for semantic object recognition (ImageNet). On subtle 128x128 ELA noise maps, its excessive depth over-analyzes routine background noise, causing it to classify almost all receipts as forged (100% recall, but only 51.83% precision and 53.54% accuracy at 395.03 ms latency). *This empirically confirms the thesis hypothesis that deeper architectures are not always superior for residual noise forensics.*

3. Why Did We Evaluate 'Heavily Compressed' Receipts?

In real-world transactions, customers send receipts via **Facebook Messenger, Viber, or WhatsApp**. These platforms aggressively downscale and re-compress images to save bandwidth. Testing heavily compressed receipts proves whether the forensic model is robust against social media transmission or if fraudsters can evade detection simply by sending receipts over chat apps.

4. Anticipated Panel Questions & Winning Responses

Q1: Why did you use ELA instead of feeding raw receipt images directly to the CNN?

Answer: 'Raw receipts vary widely in background colors, themes, and screen resolutions, which causes CNNs to memorize cosmetic elements. ELA strips away visual aesthetics and exposes the underlying mathematical JPEG compression grid, allowing the models to learn tampering signatures rather than GCash branding.'

Q2: What is your decision threshold for authentic vs. forged?

Answer: 'The final layer uses a sigmoid activation function outputting a continuous probability from 0.0 to 1.0. A standardized threshold of 0.50 is used: values below 0.50 are classified as Authentic, and values equal to or greater than 0.50 are classified as Forged.'

Q3: Why is Recall more important than Precision in receipt fraud?

Answer: 'A False Negative means a forged receipt was accepted as authentic, and goods were released without payment—causing direct financial loss. High recall guarantees that fake receipts are intercepted.'